

PHYSICS MODULE 5: ADVANCE MECHANICS

PROJECTILE MOTION

Inquiry question: How can models that are used to explain projectile motion be used to analyse and make predictions?

- analyse the motion of projectiles by resolving the motion into horizontal and vertical components, making the following assumptions:
 - a constant vertical acceleration due to gravity
 - zero air resistance
- apply the modelling of projectile motion to quantitatively derive the relationships between the following variables:
 - initial velocity
 - launch angle
 - maximum height
 - time of flight
 - final velocity
 - launch height
 - horizontal range of the projectile (ACSPH099)
- conduct a practical investigation to collect primary data in order to validate the relationships derived above.
- solve problems, create models and make quantitative predictions by applying the equations of motion relationships for uniformly accelerated and constant rectilinear motion  

CIRCULAR MOTION

Inquiry question: Why do objects move in circles?

- conduct investigations to explain and evaluate, for objects executing uniform circular motion, the relationships that exist between:
 - centripetal force
 - mass
 - speed
 - radius
- analyse the forces acting on an object executing uniform circular motion in a variety of situations, for example:
 - cars moving around horizontal circular bends
 - a mass on a string
 - objects on banked tracks (ACSPH100)  
- solve problems, model and make quantitative predictions about objects executing uniform circular motion in a variety of situations, using the following relationships:
 - $a_c = \frac{v^2}{r}$
 - $v = \frac{2\pi r}{T}$
 - $F_c = \frac{mv^2}{r}$  

$$\omega = \frac{\Delta\theta}{t}$$

- investigate the relationship between the total energy and work done on an object executing uniform circular motion

- investigate the relationship between the rotation of mechanical systems and the applied torque

$$\tau = r_{\perp} F = r F \sin \theta$$

MOTION IN GRAVITATIONAL FIELDS

Inquiry question: How does the force of gravity determine the motion of planets and satellites?

- apply qualitatively and quantitatively Newton's Law of Universal Gravitation to:

- determine the force of gravity between two objects $F = \frac{GMm}{r^2}$

- investigate the factors that affect the gravitational field strength $g = \frac{GM}{r^2}$

- predict the gravitational field strength at any point in a gravitational field, including at the surface of a planet (ACSPH094, ACSPH095, ACSPH097)

- investigate the orbital motion of planets and artificial satellites when applying the relationships between the following quantities:   

- gravitational force

- centripetal force

- centripetal acceleration

- mass

- orbital radius

- orbital velocity

- orbital period

- predict quantitatively the orbital properties of planets and satellites in a variety of situations, including near the Earth and geostationary orbits, and relate these to their uses (ACSPH101)  

- investigate the relationship of Kepler's Laws of Planetary Motion to the forces acting on, and the total energy of, planets in circular and non-circular orbits using: (ACSPH101)

$$v = \frac{2\pi r}{T}$$

$$\frac{r^3}{T^2} = \frac{GM}{4\pi^2}$$

- derive quantitatively and apply the concepts of gravitational force and gravitational potential energy in radial gravitational fields to a variety of situations, including but not limited to:  

- the concept of escape velocity $v_{esc} = \sqrt{\frac{2GM}{r}}$

- total potential energy of a planet or satellite in its orbit $U = -\frac{GMm}{r}$

- total energy of a planet or satellite in its orbit $U + K = -\frac{GMm}{2r}$

- energy changes that occur when satellites move between orbits (ACSPH096)

- Kepler's Laws of Planetary Motion (ACSPH101)

PROJECTILE MOTION

OUTCOMES

- analyse the motion of projectiles by resolving the motion into horizontal and vertical components, making the following assumptions:
 - a constant vertical acceleration due to gravity
 - zero air resistance
- apply the modelling of projectile motion to quantitatively derive the relationships between the following variables:
 - initial velocity
 - launch angle
 - maximum height
 - time of flight
 - final velocity
 - launch height
 - horizontal range of the projectile (ACSPH099)

PROJECTILE MOTION

An object moving **freely** under the **force of gravity** is called a **projectile**.

- There must be **no other force** other than the **gravitational force** acting on the object. Hence it moves “freely under the force of gravity.”
- The net force of the projectile is the gravitational force (weight).
- Objects experience:
 - o A vertically downward force of 9.8 N kg^{-1}
 - o A vertically downwards acceleration of 9.8 ms^{-2}

ANALYSING PROJECTILE MOTION

Projectile motion is **two-dimensional motion**:

- Break up the two-dimensional motion into **two one-dimensional motions**
- The horizontal motion (along x) and vertical motion (along y) are **independent** of each other (neither motion affects the other) so can be treated separately.
- There is **no acceleration** in the **horizontal direction**, as gravity only acts vertically.

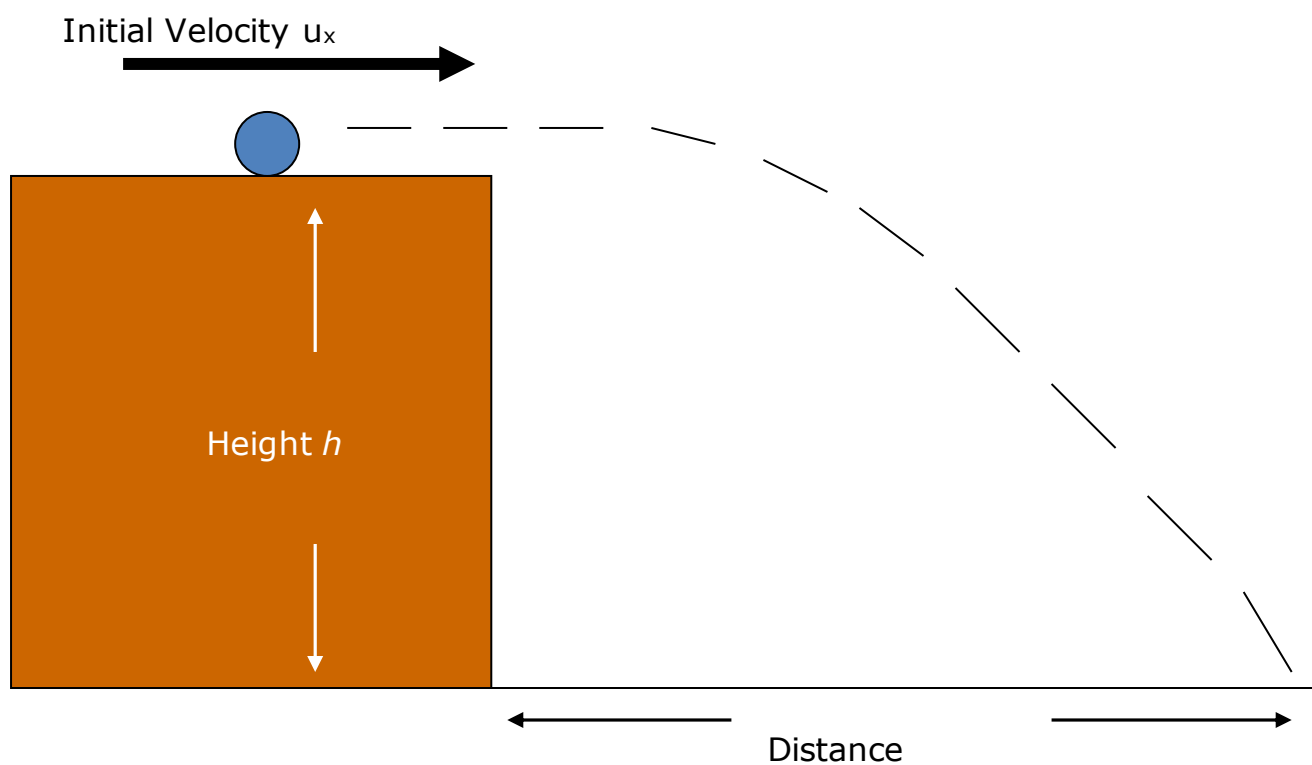
Horizontal Direction (x)	Vertical Direction (y)
$v_x = u_x$	$v_y^2 = u_y^2 + 2a_y\Delta y$
$\Delta x = u_x t$	$\Delta y = u_y t + \frac{1}{2}a_y t^2$
	$v_y = u_y + a_y t$

The following **assumptions** are made:

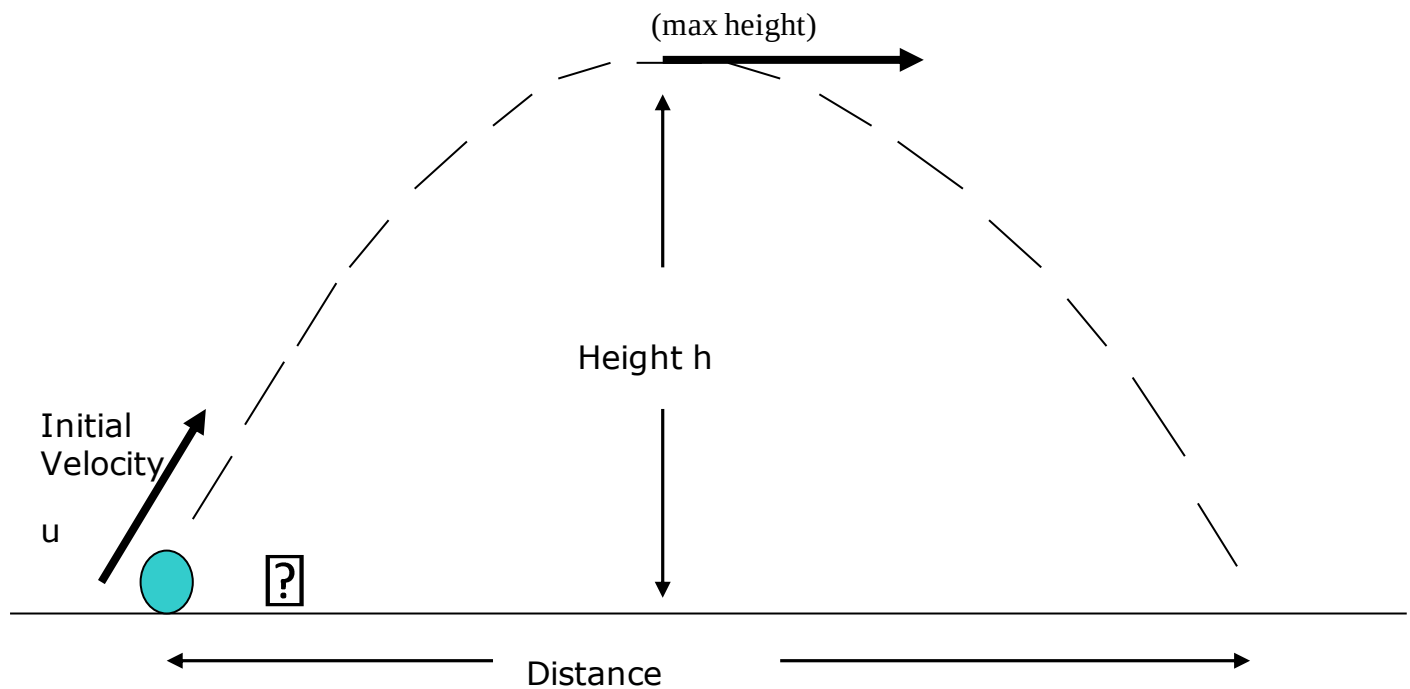
- The curvature of the Earth is ignored ($\Delta x \ll r_E$).
- The **frictional resistance** of the **atmosphere** is assumed to be negligible and ignored.

TYPES OF PROJECTILE MOTION QUESTIONS

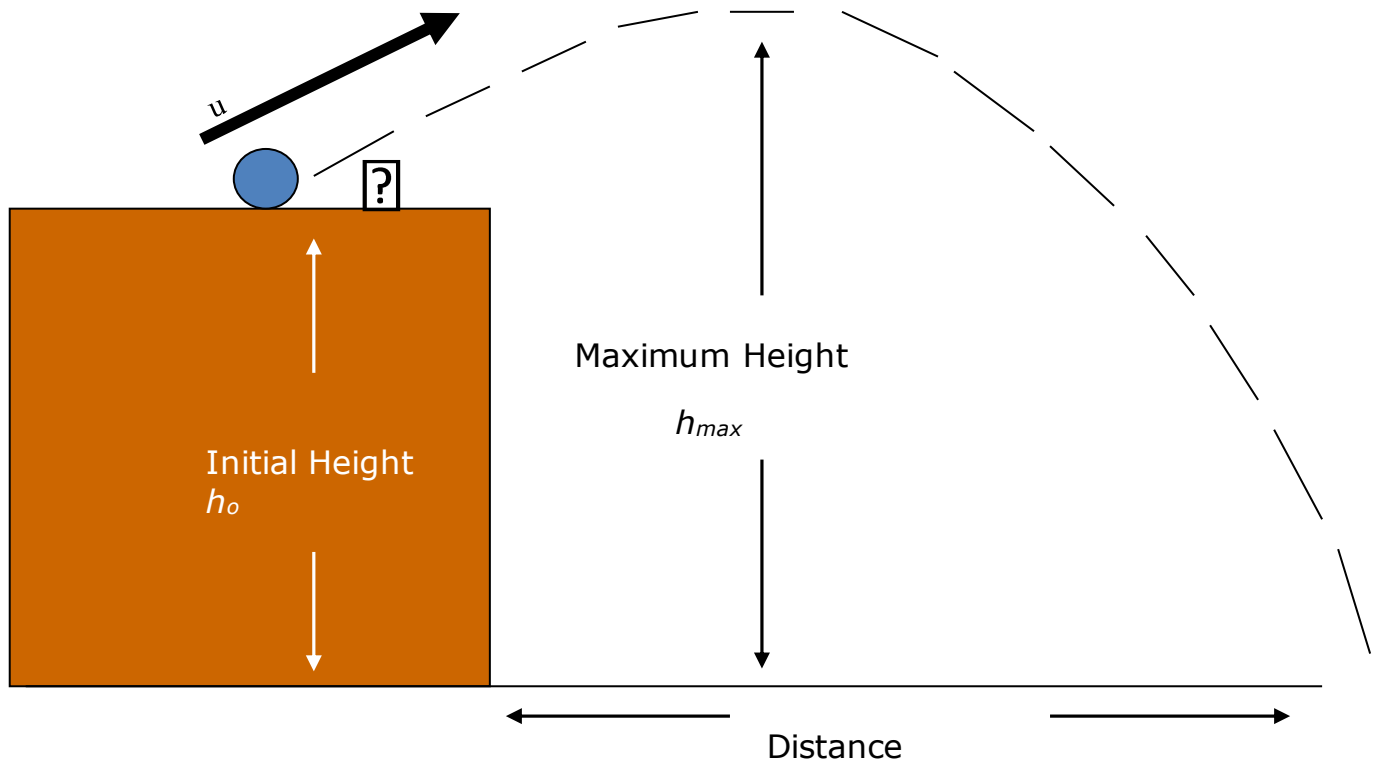
NO INITIAL VERTICAL COMPONENT



- Velocity in the x-direction: Remains at its initial value
- Velocity in the y-direction: **Initially zero** but increases as object falls.



Characteristics	Description
Initial horizontal velocity	$u_x = u \cos \theta$
Initial vertical velocity	$u_y = u \sin \theta$
Horizontal velocity at max. height	$v_x = u_x$
Vertical velocity at max. height	0
Angle of projection for max. range	45°



Characteristics	Description
Initial velocity in the x-direction	$u_x = u \cos \theta$
Initial velocity in the y-direction	$u_y = u \sin \theta$
Vertical velocity at max. height	0
Angle of projection for max range	45°

CHANGING THE ANGLE OF LAUNCH

- $45^\circ \rightarrow$ Best Horizontal and vertical distance
- $\downarrow \theta \rightarrow \downarrow \text{max height} \rightarrow \downarrow \text{time} \rightarrow \uparrow \text{Range}$
- $\uparrow \theta \rightarrow \uparrow \text{max height} \rightarrow \uparrow \text{time} \rightarrow \downarrow \text{Range}$

ROTATIONAL MOTION

OUTCOMES

- conduct investigations to explain and evaluate, for objects executing uniform circular motion, the relationships that exist between:
 - centripetal force
 - mass
 - speed
 - radius
- analyse the forces acting on an object executing uniform circular motion in a variety of situations, for example:
 - cars moving around horizontal circular bends
 - a mass on a string
 - objects on banked tracks (ACSPH100)  
- solve problems, model and make quantitative predictions about objects executing uniform circular motion in a variety of situations, using the following relationships:
 - $a_c = \frac{v^2}{r}$
 - $v = \frac{2\pi r}{T}$
 - $F_c = \frac{mv^2}{r}$  
 - $\omega = \frac{\Delta\theta}{t}$

UNIFORM CIRCULAR MOTION (UCM)

An object travelling in a **circle at constant speed** is said to be undergoing **uniform circular motion**.

The **net force** of an object moving in uniform circular motion is **directed** towards the **centre of the rotation** and is referred to as **centripetal force**.

Characteristics of uniform circular motion are:

- 1) Motion along a circular path of radius r .
- 2) Speed v is constant, hence **period** T of rotation is constant.
- 3) Angular velocity ω is constant. Angular velocity is the rate at which angular displacement changes (how quickly the angle changes).
- 4) **Linear velocity** $\vec{v}$ is **not constant** since direction is **continuously changing**. The object's linear velocity is perpendicular to the net force throughout the object's rotation.
- 5) Acceleration a is directed towards the centre of the circle (centripetal acceleration a_c).
- 6) Requires a net force (a centripetal force F_c) towards the centre of the circle.

FUNDAMENTALS OF CIRCULAR MOTION

On formula sheet

	Definition	Unit	Formula
Period T	Time taken to complete one revolution.	Seconds	$T = \frac{1}{f}$
Frequency f	The number of revolutions completed by an object in one second	Hertz	$f = \frac{1}{T}$
Angular Displacement θ	The position of the object around the circle, measured as an angle	Radians	$\frac{\theta}{180} \times \pi$
Angular velocity ω	The rate of change of angular displacement. It measures how fast or slow an object undergoes a change in angular displacement.	rad/s	$\omega = \frac{\Delta\theta}{\Delta t}$ $\omega = 2\pi f$
Linear velocity v	The rate of change of linear displacement. It measures how fast or slow the linear displacement is changing with respect to time.	ms^{-1}	$v = \frac{s}{t} = \frac{2\pi r}{T}$ $v = 2\pi r f = r\omega$
Centripetal acceleration a_c	The acceleration acting towards the centre of rotation on an object moving in a circle.	ms^{-2}	$a_c = \frac{v^2}{r}$ $a_c = r\omega^2$
Centripetal force F_c	The force acting on an object towards the centre of rotation required in order for the object to move in a circle.	N	$\sum F = ma$ $F_c = \frac{mv^2}{r}$ $F_c = mr\omega^2$

CONDITIONS FOR AN OBJECT TO EXECUTE CIRCULAR MOTION

Since the object will be accelerating, there must be a **net force** acting on the object. According to Newton's Second Law:

$$\sum F = ma$$

$$F_{net} = ma$$

In **uniform circular motion**, the **net force** is directed towards the **centre** of the circular path and is referred to as the **centripetal force** F_c :

$$F_{net} = F_c = \frac{mv^2}{r}$$

Situation	Force providing the centripetal force	Condition
Car driving around a corner	Friction between the tyres and the road f	$f = \frac{mv^2}{r}$
Ball swinging around on a string	Tension in the string T	$T = \frac{mv^2}{r}$
Satellite orbiting a planet	Gravitational force between the satellite and the planet F_G	$F_G = \frac{mv^2}{r}$

If the centripetal force and the details of the motion do not satisfy $F_c = \frac{mv^2}{r}$, then the object will not execute uniform circular motion. It will not follow a circular path and instead follow a different trajectory.

Situation	Condition	Motion
Car driving around a corner	The road is slippery, there is not enough friction $F < \frac{mv^2}{r}$	The car will slide out of the turn into a greater radius
Ball swinging around on a string	The string is cut so tension in the string $T = 0 < \frac{mv^2}{r}$	The ball travels in a straight line.
Satellite orbiting a planet	The satellite's motion does not satisfy the conditions of circular motion, hence $F_G \neq \frac{mv^2}{r}$	The satellite will move into an elliptical orbit.

RPM $\rightarrow$ rad/s

Revolutions per minute or 'RPM' is commonly used in situations involving circular motion.

Conversion of RPM to rad/s:

- 1) Convert the minutes to per second by dividing by 60.
- 2) Convert the revolutions to radians by multiplying by 2π

PROBLEMS INVOLVING UNIFORM CIRCULAR MOTION

HOW TO ANALYSE UCM QUESTIONS

- 1) **Identify the forces** acting on the object. Draw a **free body diagram**.
- 2) Determine the **direction of acceleration**. For UCM, the acceleration is always **towards the centre of the circular path**.
- 3) **Decompose the forces** so they are either **parallel to the acceleration** or **perpendicular to it**.
- 4) **Apply Newton's Second Law** to the direction of acceleration and the direction perpendicular to that, and find the **unknowns**:
 - In the direction of acceleration there must be a net force: $F_{net} = \frac{mv^2}{r}$
 - In the direction perpendicular to the acceleration $a = 0$ so $F_{net} = 0$
 - A **common unknown** that must be solved is **centripetal acceleration**. Once you know the acceleration, other quantities can be found, such as unknown forces or the speed.

VEHICLES ROUNDING A CURVE ON FLAT ROAD

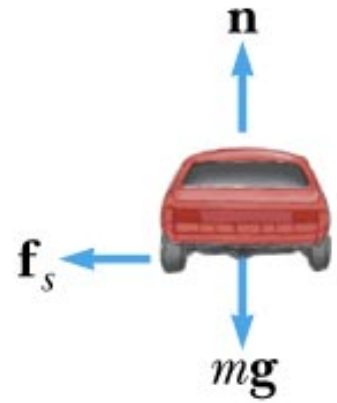
- 1) Lateral (perpendicular) force between tyres and the road surface (f)
- 2) Normal force (N)
- 3) Weight force (w)

The car does not accelerate vertically. Hence:

$$\Sigma F_y = ma_y$$

$$N - w = 0$$

$$\therefore N = w$$



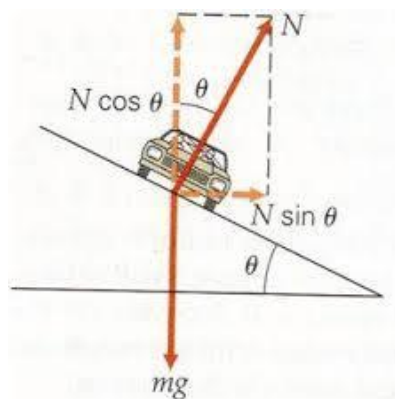
The driver can change the amount of lateral frictional force by **turning the steering wheel**, which causes the front wheels to push the car left or right. The following situations are possible:

Centripetal Force	Motion of Car
$F = \frac{mv^2}{r}$	Uniform Circular Motion
$F > \frac{mv^2}{r}$	Car will move towards the centre of the circle . Radius creases so the car turns more sharply.
$F < \frac{mv^2}{r}$	Car will move away from the centre of the circle . Radius increases so car turns more gently. This will also happen if there is not enough friction due to the road being slippery.

CAR ON BANKED TRACK WITH NO FRICTION

On a **cambered** or **banked road**, the **road is tilted into the centre of the turn** or circular path. Banked tracks allow vehicles to travel at higher speed without skidding.

- Thus the **normal force is tilted** and has a component towards the centre of the circle.
- **The normal force then contributes to the centripetal force.**
- If the car is turning at the correct speed, the **horizontal component** of the **normal force** will provide the centripetal force and the car turns in a circle **without relying on friction**.



$$\Sigma F_y = N \cos \theta - mg = 0$$

$$\therefore N \cos \theta = mg \quad (1)$$

$$\Sigma F_x = N_x = ma_x$$

$$\therefore N \sin \theta = ma_x \quad (2)$$

$$(1)/(2)$$

$$\frac{N \sin \theta}{N \cos \theta} = \frac{ma}{mg}$$

$$\tan \theta = \frac{a}{g}$$

$$a = g \tan \theta$$

$$F_c = mg \tan \theta$$

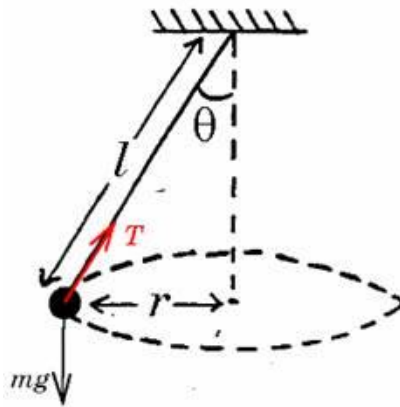
$$a = g \tan \theta = \frac{v^2}{r}$$

$$\therefore v = \sqrt{gr \tan \theta}$$

CONICAL PENDULUM

A pendulum is a mass m at the end of a light string of length l . The top end is fixed to a frame or mount.

The forces on the mass are **weight force** downwards and **tension force** in the direction of the string. The string is at an angle of θ from vertical. The pendulum rotates with a radius r , at a distance h below the mount.



The analysis of the forces on the conical pendulum gives:

- The vertical component of tension $T_y = T \cos \theta$ balances the weight force:

$$\Sigma F_y = T \cos \theta - mg = 0$$

$$T \cos \theta = mg$$

$$T = \frac{mg}{\cos \theta}$$

- The horizontal component of tension $T_x = T \sin \theta$ provides the centripetal force, using $T = \frac{mg}{\cos \theta}$

$$\Sigma F_x = T \sin \theta = ma$$

$$\Sigma F_x = mg \tan \theta = \frac{mv^2}{r}$$

- The horizontal component of the tension **provides the centripetal force** for the mass on the pendulum.

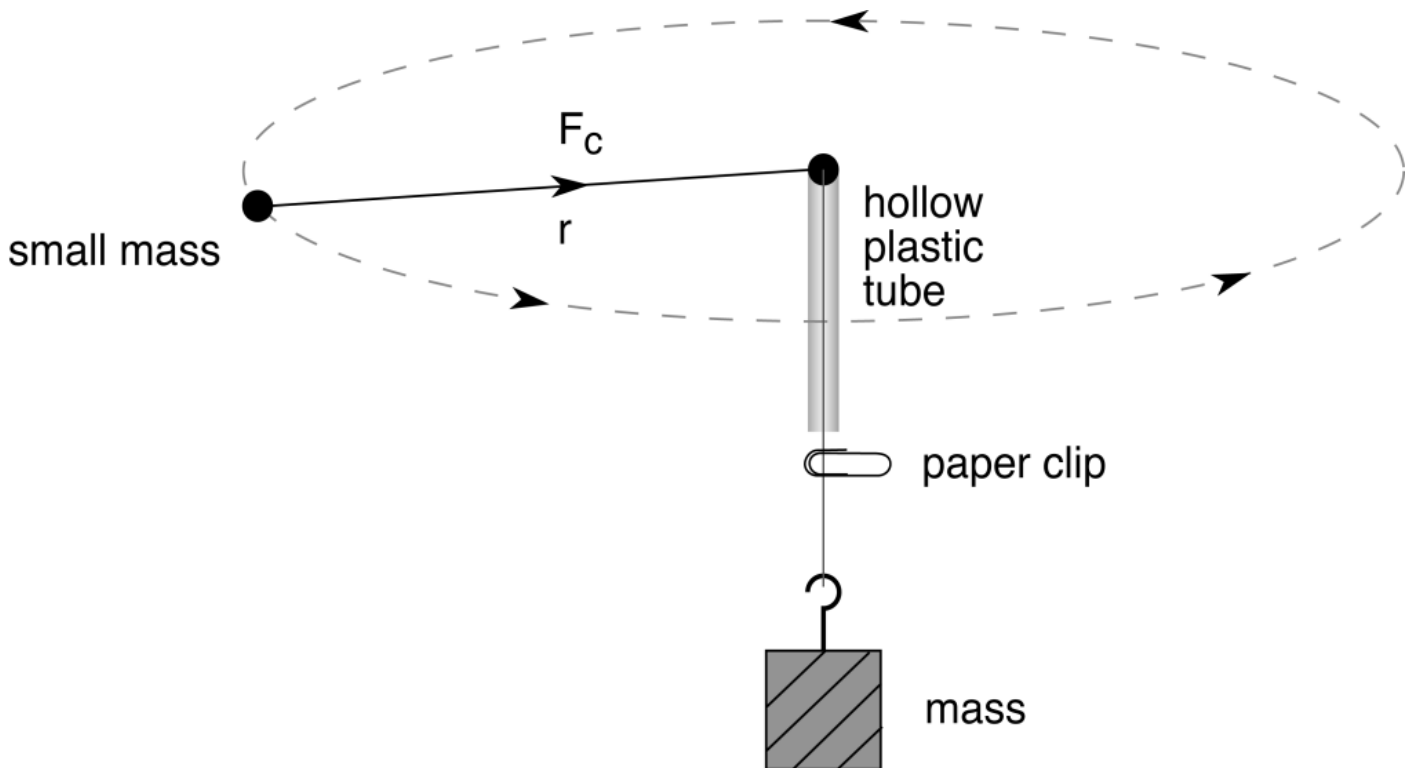
INCREASE IN TENSION ON CONICAL PENDULUM

- Vertical component stays **constant** → Balances the downward weight force → Angle increases
- Horizontal component **increases** → Centripetal force **increases** → Angle increases

CENTRIPETAL MASS BALANCE

A centripetal mass balance has a mass m spinning on a string. The other end of the string is attached to a hanging mass M and allowed to slide through a frictionless tube or pulley.

- The mass m is spun at a certain speed so that the mass M remains stationary.



$$\text{On } M: \Sigma F = T - Mg = 0$$

$$\text{On } M: \text{Therefore } T = Mg$$

$$\text{On } m: T \cos \theta = mg$$

$$\text{On } m: \text{Therefore } T = \frac{mg}{\cos \theta}$$

$$Mg = \frac{mg}{\cos \theta}$$

$$M = \frac{m}{\cos \theta}$$

Relationship between masses and the angle at which it must rotate: $\frac{m}{M} = \cos\theta$

Equation for Centripetal Force:

$$T_x = T \sin\theta = ma$$

$$T_y = T \cos\theta = mg$$

$$\tan\theta = \frac{a}{g}$$

$$a = g \tan\theta$$

$$a = g \tan(\cos^{-1}(\frac{m}{M}))$$

$$F_c = mg \tan(\cos^{-1}(\frac{m}{M}))$$

OUTCOMES

- investigate the relationship between the total energy and work done on an object executing uniform circular motion

UNIFORM CIRCULAR MOTION OF RIGID BODIES

A **rigid body** is a **solid object** or structure. Eg:

- Discs like wheels on a car or bicycle, or a ferris wheel.
- Spheres like a ball or the Earth.

When a rigid body rotates the **entire object moves together**.

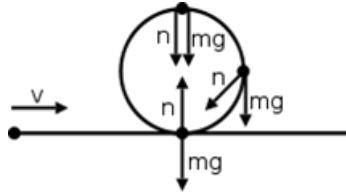
- All parts of the object undergo uniform circular motion.
- The **angular velocity** ω is the **same** for **all parts** of the object.
- The **linear velocity** $v = r\omega$ **depends on position**.

NON UNIFORM CIRCULAR MOTION

THE LOOP-DE-LOOP MANEUVER

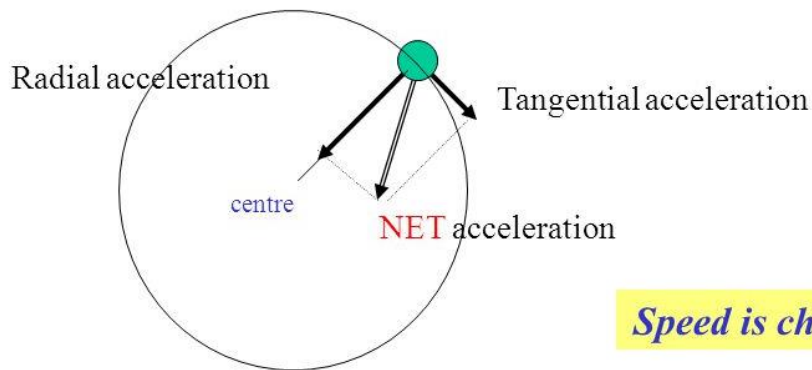
In a **loop-de-loop**, a vehicle gains speed during a run up, and enters a vertical circular track that takes the vehicle upside down at the top of the circle, to upright again at the bottom of the circle.

- In a loop-de-loop, a vehicle undergoes **non-uniform circular motion** because the **speed changes** around the circle, unlike a rigid body.
- By analysing the changes in kinetic and potential energy, it can be determined that it is non-uniform



At any instant the object is still in circular motion so:

- Its **acceleration** is $a_c = \frac{v^2}{r}$. The acceleration **changes** around the loop since the speed changes.
- The force is $F = ma = \frac{mv^2}{r}$. The force **changes** around the loop.
- The **acceleration is towards the centre**.
- The **radial direction is towards the centre**, since a vector pointing towards the centre would be along a radius of the circle.
- The **tangential direction** is perpendicular to the radial since a tangent to a circle is perpendicular to the radius.



Speed is changing

Position	Radial Force (Positive towards the centre)	Tangential Force (Positive in the same direction as velocity)
A: Bottom (Lowest Point)	$N - mg$	0
B: Right	N	$-mg$
C: Top (Highest Point)	$N + mg$	0
D: Left	N	$+mg$

At B: **Weight** is **antiparallel** to the direction of velocity and therefore is doing **negative work**.

At D: **Weight** is **parallel** to the direction of velocity, therefore is doing **positive work**.

Centripetal Force at each point:

- A: $N - mg = \frac{mv^2}{r}$
- B & D: $N = \frac{mv^2}{r}$
- C: $N + mg = \frac{mv^2}{r}$

The minimum value is when $N = 0$.

TORQUE

OUTCOMES

- investigate the relationship between the rotation of mechanical systems and the applied torque
 - $\tau = r_{\perp} F = r F \sin \theta$

DEFINITION OF TORQUE

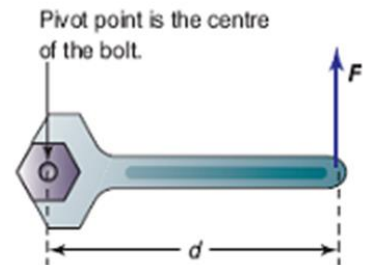
Torque is defined as the **tendency of a force to rotate** an object about an **axis**.

The **magnitude** of the torque is calculated by the following formula:

$$\text{Torque} = \text{Perpendicular distance} \times \text{Force}$$

$$\tau = r_{\perp} F = r F \sin \theta$$

- τ : Torque (Nm)
- $r_{\perp}$: The perpendicular distance from the axis of rotation to the force.



The **pivot point** is the **centre of the bolt**. Since torque is a **vector**, the direction of torque is usually stated as being either **clockwise** or **anticlockwise**.

CONDITIONS FOR MAXIMUM TORQUE

For a given force F , the **torque** produced by the forces is **maximum** when the force applied is **perpendicular** to the line joining the forces and the pivot point.

EFFECT OF TORQUE ON THE ROTATION OF A SYSTEM

Torque is the equivalent force for rotational motion:

- Force will cause linear acceleration in linear motion
 - $\frac{\Delta v}{t} \propto F$
- Torque** will cause **angular acceleration** in rotational motion
 - $\frac{\Delta \omega}{t} \propto \tau$

Torque is applied whenever there is a force acting **tangentially** to rotational motion.

- Torque will **increase angular velocity** if the **tangential component of the force** is in the **same direction as the velocity**.
- Torque will **decrease angular velocity** if the **tangential component of the force** is in **opposite directions to the velocity**.

TORQUE OR ROTATIONAL EQUILIBRIUM

An object in **rotational equilibrium** has no net external torque. This however does not mean that the object is not rotating, it may seem that the object is rotating with constant angular velocity.

GRAVITATIONAL FIELDS

OUTCOMES

- apply qualitatively and quantitatively Newton's Law of Universal Gravitation to:
 - determine the force of gravity between two objects $F = \frac{GMm}{r^2}$
 - investigate the factors that affect the gravitational field strength $g = \frac{GM}{r^2}$
 - predict the gravitational field strength at any point in a gravitational field, including at the surface of a planet (ACSPH094, ACSPH095, ACSPH097)
 - derive quantitatively and apply the concepts of gravitational force and gravitational potential energy in radial gravitational fields to a variety of situations, including but not limited to:
 - total potential energy of a planet or satellite in its orbit $U = -\frac{GMm}{r}$
 - energy changes that occur when satellites move between orbits (ACSPH096)

COMPARISON OF MASS AND WEIGHT

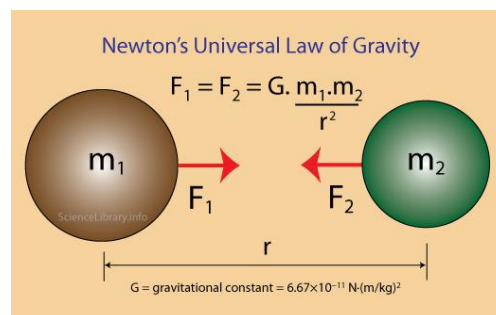
	Mass	Weight
Definition	Mass is the measure of the inertia an object exhibits	Weight is the force with which an object is attracted to the Earth due to gravity.
SI Units	<i>kg</i> (kilogram)	<i>N</i> (Newton)

GRAVITATIONAL FORCE OF ATTRACTION

NEWTON'S LAWS OF ATTRACTION

The existence of weight can be explained by the **gravitational force of attraction** which acts on **all objects** with **mass**. The Law of Universal Gravitation states that:

“Every object (mass) in the universe attracts each other object (mass) with a force directly proportional to the product of their masses and inversely proportional to the square of the distance from their centres”



$$F = \frac{Gm_A m_B}{r^2}$$

- m_A and m_B are the masses of any two objects
- r is the distance between their centres
- G is the universal gravitational constant $6.67 \times 10^{-11} \text{ Nm}^2 \text{ kg}^{-2}$

GRAVITATIONAL FIELD

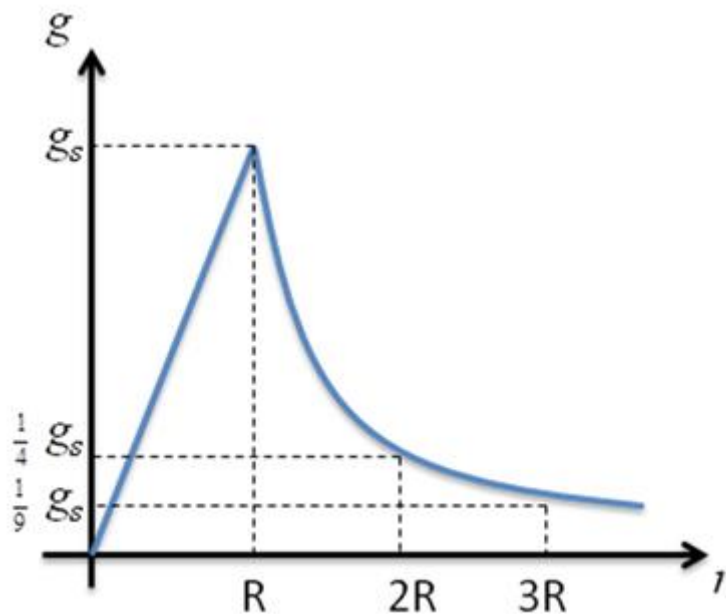
The **gravitational field** is a model used to describe the **region of influence** within which the **gravitational force** between any two objects with **mass** exists.

- The gravitational field is used to describe gravitational phenomena and is measured in Newtons per kilogram (Nkg^{-1})
- Gravitational phenomena: Mass $\rightarrow$ Gravitational field $\rightarrow$ Gravitational force of attraction experience by another mass in the field $\rightarrow$ Weight and acceleration due to gravity

The **Earth's gravitational field strength** at a distance from the centre of the earth is: $g_E = \frac{GM_E}{r^2}$

- M_E = mass of the Earth
- r = distance from the centre of the earth
- r_E = radius of the earth (6371 km)

GRAVITATIONAL FIELD VS. DISTANCE FROM CENTRE OF THE EARTH



ACCELERATION DUE TO GRAVITY

The acceleration due to gravity is **equal in magnitude** to the gravitational field strength.

- If gravity is the only force acting on an object, then it will be the net force $F_{net} = mg$
- From Newton's Second Law $F_{net} = ma$
- Therefore $a = g$

The **acceleration due to gravity** is the **same for all objects at the same point** in a gravitational field.

GRAVITATIONAL POTENTIAL ENERGY

DEFINITION OF WORK

The **work done** on an object is defined as the **force acting** on the object over a **displacement**.

- It is mathematically given by: $W = Fs \cos \theta$
- It is only the component of the force in the **direction of displacement** that does work on an object.

If F is the net force, then the object will accelerate, and the velocity changes. **Work causes a change of the kinetic energy** of the object.

POTENTIAL ENERGY

$$U = - \frac{GMm}{r}$$

This defines potential energy as the **work done** by an **upward external force** on an object of mass m as it is **lowered from one position to another without acceleration**.

- Specifically, this defines the gravitational potential energy as the work done in moving an object **from an infinite distance** to a **distance r** from a centre of a large mass M .

COMPARISON OF POTENTIAL ENERGY

	$U = mgh$	$U = - \frac{GMm}{r}$
Definition	The work done against the gravitational field in moving a mass from the surface of the Earth to a position of height h above.	The work done by an upward external force in lowering a mass from infinity to a distance r from the centre of the Earth without acceleration.
$U = 0$	At the surface of the Earth (arbitrary definition)	At infinite distance away from the centre of the Earth
As r increases	U increases to infinity	U increases to 0 at infinity
Validity	Formula valid near the Earth's surface $h \ll r_E$ so that g is approximately constant	Formula valid for all $r > r_E$. Used for large height changes or far from Earth's surface

CHANGES IN GRAVITATIONAL POTENTIAL ENERGY

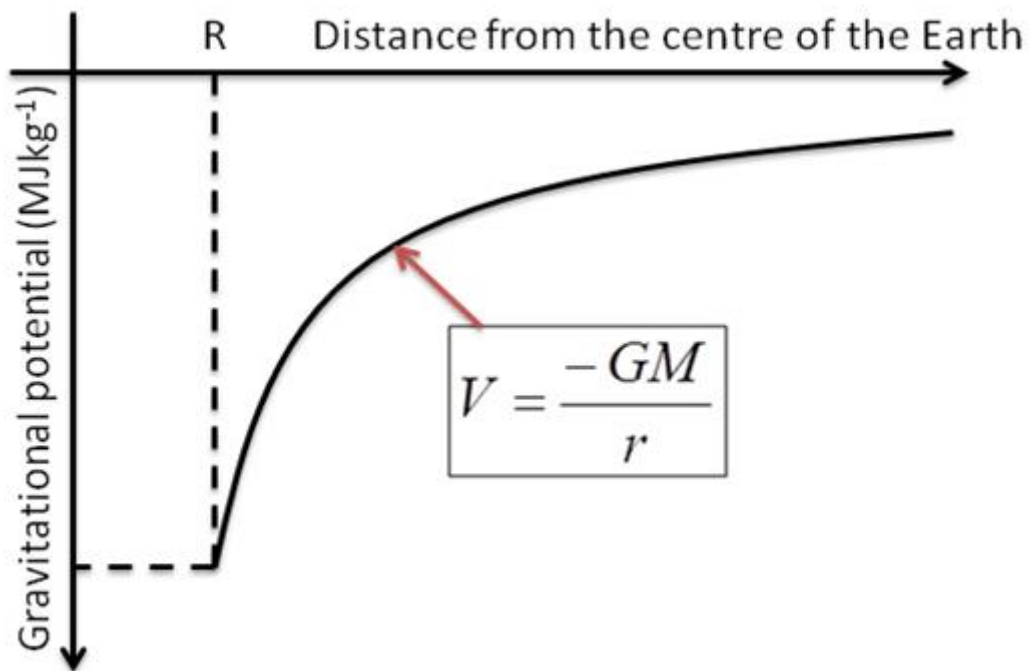
When the altitude **decreases**, gravitational potential energy **decreases**. Therefore the change in gravitational potential energy is negative.

When the altitude **increases**, gravitational potential energy **increases**. Therefore the change in gravitational potential energy is positive.

$$\Delta U = U_f - U_i$$

$$\Delta U = - GMm \left(\frac{1}{r_f} - \frac{1}{r_i} \right)$$

Note: It is important to add all distances.



DERIVATION

$$\begin{aligned}
 & \int_{\infty}^r \vec{F} dr \\
 &= \int_{\infty}^r -\frac{GMm}{r^2} dr \\
 &= -GMm \int_{\infty}^r \frac{1}{r^2} dr \\
 &= -GMm \left[\frac{1}{r} \right]_{\infty}^r \\
 &= -\frac{GMm}{r}
 \end{aligned}$$

ORBITS

OUTCOMES

- investigate the orbital motion of planets and artificial satellites when applying the relationships between the following quantities:   
 - gravitational force
 - centripetal force
 - centripetal acceleration
 - mass
 - orbital radius
 - orbital velocity
 - orbital period
- predict quantitatively the orbital properties of planets and satellites in a variety of situations, including near the Earth and geostationary orbits, and relate these to their uses (ACSPH101)  

ORBITS OF SATELLITES

A satellite is an object that **moves around another larger object**. There are two types of satellites:

- Natural satellites, eg. moon
- Artificial satellites eg. International space station

ORBITAL SPEED OF SATELLITES

The Earth's **gravitational field holds the satellite** in a certain path as it goes around the Earth, and that path is called an **orbit**. In a stable orbit:

- The radius of the orbit remains constant.
- The **gravitational force of attraction** between the Earth and the satellite acts as the **centripetal force** required to execute uniform circular motion around the **Earth**.

$$F_G = \frac{Gm_s M_E}{r^2}$$

$$v = \frac{2\pi r}{T} = \sqrt{\frac{GM_E}{r}}$$

There are two methods for calculating **orbital speed** of a satellite:

DERIVATION OF ORBITAL SPEED

Gravitational force of attraction = Centripetal Force

$$F_G = F_c$$

$$\frac{GM_E m_s}{r^2} = \frac{m_s v^2}{r}$$

$$v^2 = \frac{GM_E}{r}$$

$$v = \sqrt{\frac{GM_E}{r}}$$

The orbits of satellites are described in two different ways:

- By indicating the **radius of the orbit r** . This is the distance between the Earth and the satellite.
- By indicating the **altitude of the orbit h** . This is the height of the orbit above the ground.
- The equation $v = \sqrt{\frac{GM}{r}}$ uses the **radius** of the orbit, not the altitude.

CENTRIPETAL ACCELERATION

An object in orbit will accelerate towards the Earth at the acceleration due to gravity.

- The object will have a **tangential velocity**.
- The **acceleration towards the ground** is what **keeps the object in uniform circular motion**.

NEWTON'S THOUGHT EXPERIMENT

Newton predicted that an object projected horizontally from a high mountain would undergo projectile motion.

- He suggested that a ball could be pushed forward and it would travel a short distance before reaching the ground. If it were pushed **harder**, it would travel further before touching the ground.
- If pushed fast enough, it **would fall towards the Earth at the same rate as the Earth curves away from it**. It would be falling towards the Earth, but its path would be parallel to the Earth's surface, so its height above the ground would remain constant. **This is circular orbit**.
- The ball would constantly fall towards the ground, it just wouldn't get closer to the ground.

WEIGHTLESSNESS

APPARENT WEIGHT AND TRUE WEIGHT

The **true weight** of a person is **always their actual weight** (mg), the gravitational force of attraction to the Earth. **True weight does not change.**

The **apparent weight** of a person is the **normal force** acting on the person from the platform which they are standing on.

- Apparent weight is responsible for the **sensation of weight**.
- Since apparent weight comes from the platform supporting you, it **can vary**.

EFFECT OF ACCELERATION ON APPARENT WEIGHT

- 1) When acceleration is 0, apparent weight = true weight.
- 2) When acceleration is upward, apparent weight > true weight. (Feels heavier)
- 3) When acceleration is downward, apparent weight < true weight. (Feels lighter)

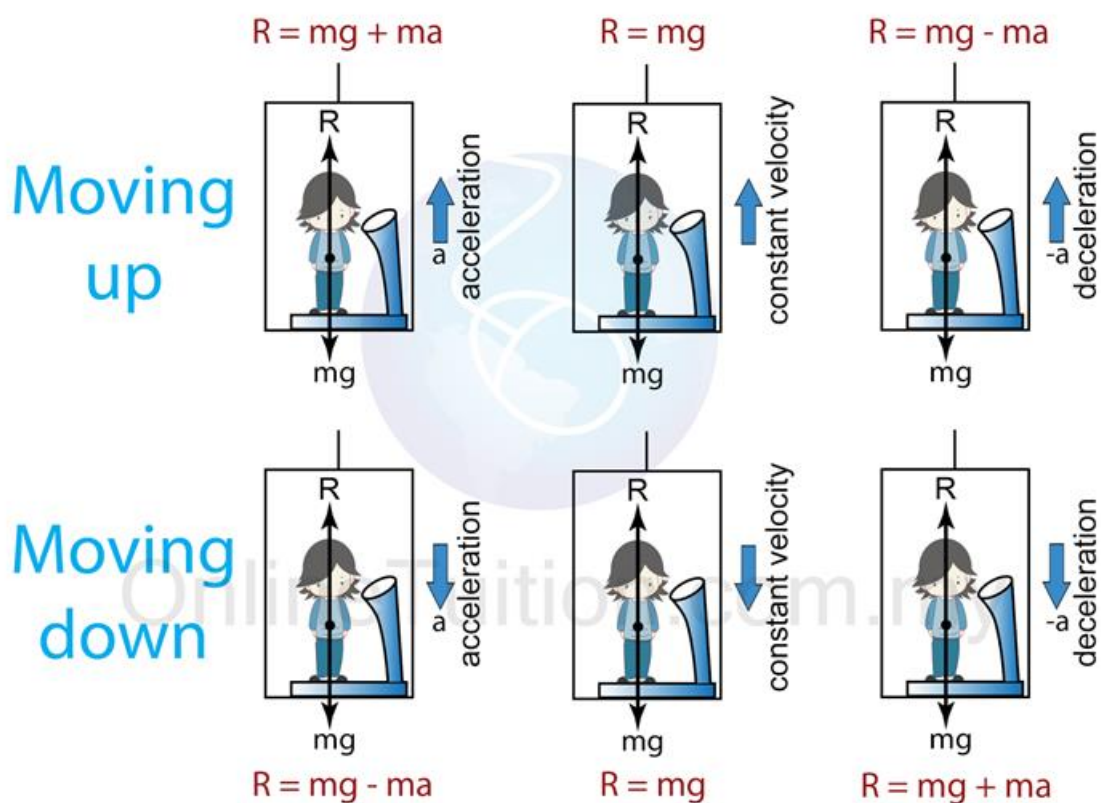
WEIGHTLESSNESS

Weightlessness occurs when there is **no force of support on your body**, hence when **apparent weight is 0**.

- When your body is in free fall, accelerating downwards at the acceleration due to gravity, then you are not being supported (there is no force acting on you other than gravity).
- **The sensation of apparent weight** comes from the **support** that you feel from the floor, platform.
- Hence, if there is no support, there is **no apparent weight**.

For example: ISS is in free fall under the influence of Earth's gravitational field → Astronauts in the ISS are also in free fall → No normal reaction force from ISS acting on astronaut → Astronaut apparent weight = 0 → Apparent weightlessness

SUMMARY



Acceleration Downwards:

$$\begin{aligned}\sum \vec{F}_y &= m\vec{a} \\ \vec{F}_g - \vec{F}_N &= m\vec{a} \\ \therefore \vec{F}_N &= \vec{F}_g - m\vec{a} \\ \therefore \vec{F}_N &< \vec{F}_g\end{aligned}$$

Meaning the 'apparent weight' is less than the actual weight.

Acceleration at acceleration due to gravity:

$$\begin{aligned}\sum \vec{F}_y &= m\vec{g} \\ \vec{F}_g - \vec{F}_N &= m\vec{g} \\ \therefore \vec{F}_N &= m\vec{g} - m\vec{g} = 0 \\ \therefore \vec{F}_N &= 0\end{aligned}$$

Meaning the 'apparent weight' is 0 and they 'apparent weightless'

TYPES OF SATELLITE ORBITS

Satellites can be positioned in orbits with different altitudes and shapes. Based on orbital radius, satellites fall into one of three categories:

- 1) LEO: Low Earth Orbit
- 2) MEO: Medium Earth Orbit
- 3) GEO: Geostationary Earth Orbit

LEO: LOW EARTH ORBIT

A Low Earth Orbit describes a satellite which orbits Earth with altitudes around 160km - 1000km. Most LEOs are **polar orbits**.

- The ISS is 460km above the Earth's surface
- The Hubble Space Telescope is 559km above the Earth's surface

LEO satellites are used for a variety of purposes:

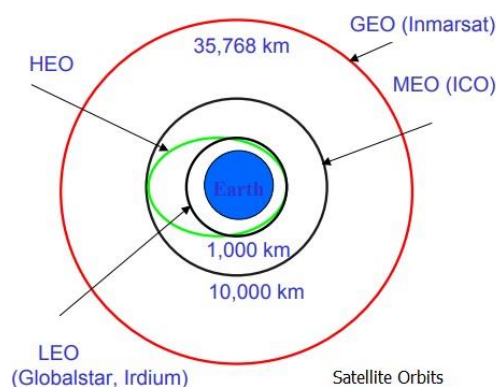
- Remote sensing to study the surface of the Earth for minerals
- Military activity
- Telecommunications applications
- Special satellites like the Hubble Space Telescope

Due to the low altitude, a small amount of atmospheric drag (air resistance) acts on LEO satellites.

- As a result, over time, LEO satellites slow down and their orbits **slowly decrease**. This is known as **orbital decay**.

Satellites in LEO can be used to provide relays for telecommunication applications. A **constellation of LEO satellites** is required in order to provide continuous coverage:

- The coverage of LEO satellites is relatively narrow due to their proximity to Earth
- The orbital speed means a satellite is only overhead and in range for a short duration
- However, LEO satellites can capture detailed images of the Earth's surface due to their low altitude and **do not require powerful amplifiers for successful transmission**.



ELLIPTICAL ORBIT

A satellite in elliptical orbit follows an **oval-shaped** path:

- One part of the orbit is closest to the center of the Earth (perigee) and another part is farthest away (apogee)
- A satellite in this type of orbit generally has an inclination angle of 64 degrees and takes 12 hours to circle the planet.
- This type of orbit covers regions of high latitude for a large fraction of its orbital period.

A **constellation of satellites** can provide **continuous coverage** in order to be used for communication purposes. In the case of the **GPS system**, there are **three or more** satellites covering **any location on the planet**.

GEOSTATIONARY ORBIT

A geostationary orbit is one in which the satellite is **always in the same position** with respect to the rotating Earth. The velocity of the satellite relative to the ground is **zero**.

The satellite orbits at an elevation of approximately 35,800km because that produces an **orbital period equal to the period of rotation of the Earth**. Hence, the satellite appears to remain in the same position above the Earth.

GEOSYNCHRONOUS ORBIT

An orbit in which the satellite completes one circuit around the Earth in one day, then appears in the same position above the Earth's surface is known as a **geosynchronous orbit**. Geostationary orbits are a special case of geosynchronous orbits.

A satellite in geosynchronous orbit goes around the Earth **once a day** before returning to its **original position**.

- The duration of **one orbit** is the same as **Earth's rotational period**.
- The **orbital inclination** is not always zero and shape of the orbit may be elliptical.
- A geostationary orbit is thus a geosynchronous orbit with **orbital inclination of zero** and a **circular shape**.

SUMMARY: LEO VS GEO

Type	LEO	GEO
Description	Low Earth Orbit	Geostationary Earth Orbit
Altitude	160 - 1000 km	35800 km
Radius	6531 - 7371 km	42300 km
Velocity	$7368 - 7828 m.s^{-1}$	$3076 m.s^{-1}$
Period	87 - 105 minutes	24 hours
Time in Line of Sight	15 minutes	24 hours
Function	- Remote sensing - Military activity - Telecommunications - Specialised satellites	- Communications - Weather forecasting
Advantages	- Relatively cheap to build and deploy - Repair work can be carried out whilst in orbit - Can capture detailed images of Earth's surface - Does not require a powerful amplifier for successful transmission	- Always remains in the same position - No tracking is required from the ground - The satellite can provide continuous operation in the area of visibility of the satellite
Disadvantages	- Small amount of atmospheric drag acts on the satellite. As a result, over time, their orbits slowly decay - Time in line of sight is very short. A constellation of satellites is required to provide continuous coverage	- Expensive to build and deploy - Repair work cannot be carried out - Cannot provide full coverage for high latitude regions (Russia)
Examples	- Hubble Space Telescope - International Space Station	- Intercontinental communication satellites

ORBITAL DECAY

All satellites in LEO are subjected to some degree of friction called **atmospheric drag**. The amount of atmospheric drag depends on:

- **Density of air**
- **Speed** of the satellite
- The **shape and size** of the satellite

KEPLER'S LAWS

OUTCOMES

- investigate the relationship of Kepler's Laws of Planetary Motion to the forces acting on, and the total energy of, planets in circular and non-circular orbits using: (ACSPH101)
 - $v = \frac{2\pi r}{T}$
 - $\frac{r^3}{T^2} = \frac{GM}{4\pi^2}$  
- derive quantitatively and apply the concepts of gravitational force and gravitational potential energy in radial gravitational fields to a variety of situations, including but not limited to:  
 - Kepler's Laws of Planetary Motion (ACSPH101)

KEPLER'S THREE LAWS - EMPIRICAL LAWS

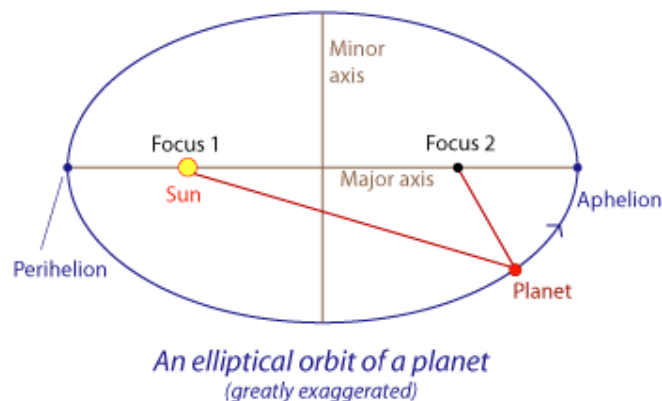
Kepler's three laws of planetary motion are all **empirical laws** based on **observation**. They only **describe** the **patterns in observation**. They do not attempt to explain them with a more general underlying theory.

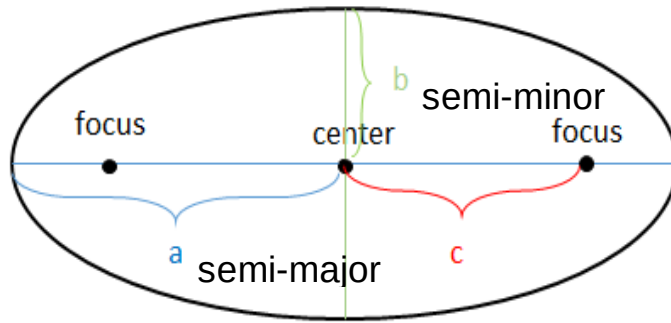
Kepler proposed that the motion of the planets around the Sun can be described by the following three laws:

- Planet move around the Sun in an elliptical orbit which has the sun in one of its two foci
- A line joining the planet to the Sun sweeps out equal areas in equal times as the planet orbits
- The radius of the orbit cubed divided by period of the orbit squared is the same for all the planets in our Solar System

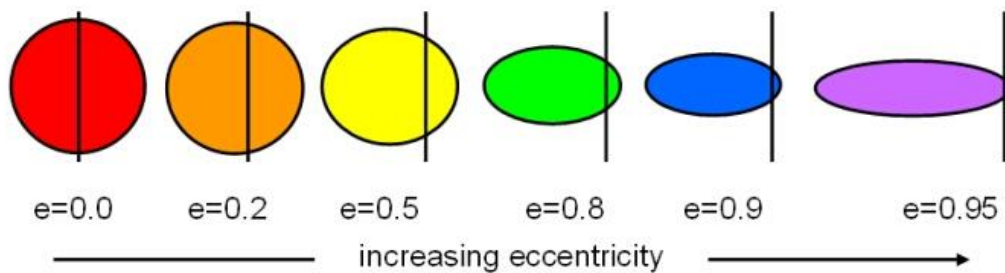
KEPLER'S FIRST LAW: LAW OF ELLIPSES

Each planet has an elliptical orbit with the Sun at one focus.





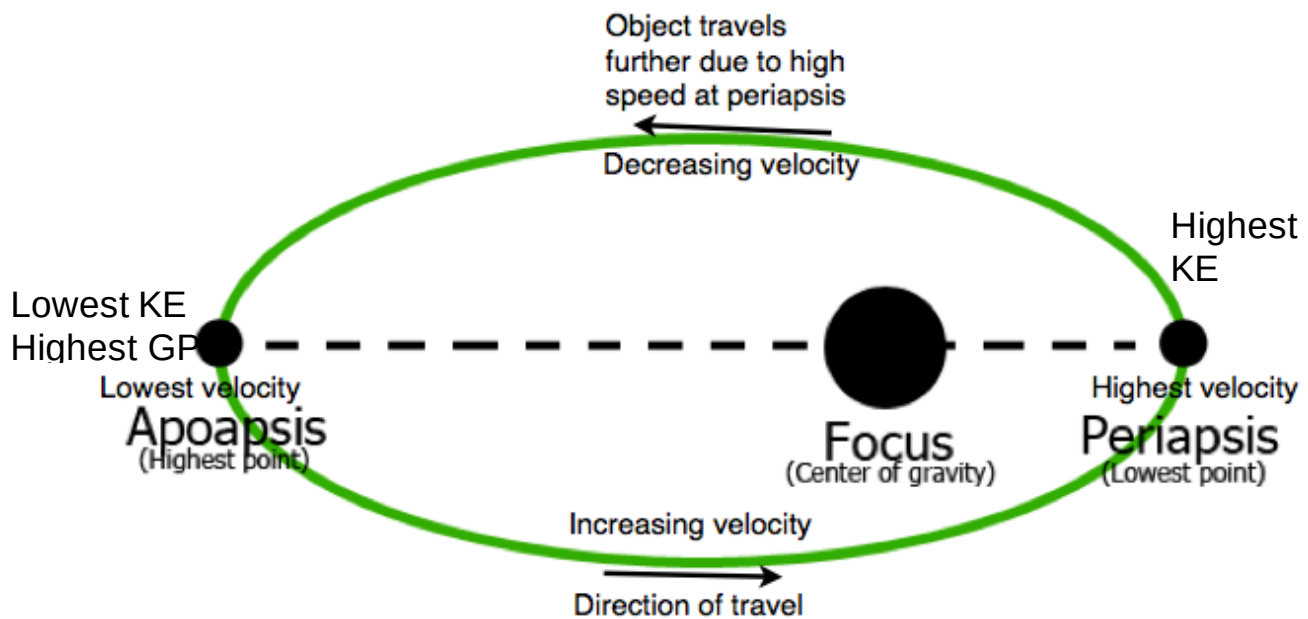
- The sum of distance to both foci is a constant
- **Eccentricity** is how flat the ellipse is. It is defined as, $e = \sqrt{1 - \frac{b^2}{a^2}}$



The orbits of all planets except for Mars, Mercury and Pluto are very close to being circular.

- The ratio of the semi-minor to the semi-major axis for the Earth is $\frac{b}{a} = 0.99986$. The eccentricity is 0.017
 - The orbit of Earth is **nearly circular**.

Elliptical orbits have the **central body** at **one focus** of the ellipse. For elliptical orbits, the central body will not be in the center of the orbit. This can also be related to Kepler's Second Law.



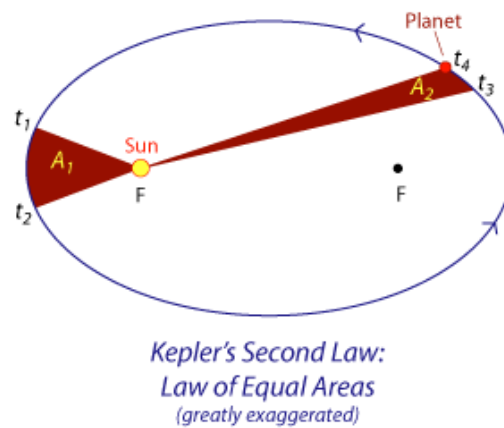
The distance between the central body and the orbiting body (satellite) will vary as satellites move around the ellipse.

- The **minimum distance** between the satellite and the central body occurs when the satellite **crosses the major axis nearest the central body**. The general name for this position is **periapsis**, but the special names are given depending on the central body.
- Similarly, the **maximum distance** is when the satellite crosses the **major axis** furthest from the central body. The general name for this position is **apoapsis**.

Central Body	Name of Closest Point	Name of Furthest Point
General	Periapsis/Pericenter	Apoapsis/Apocentre
Sun	Perihelion	Aphelion
Earth	Perigee	Apogee
Star	Periastron	Apastron

KEPLER'S SECOND LAW: LAW OF AREAS

Each planet moves so that a radius vector from the Sun to the planet sweeps out equal areas in equal times.



Derivation of Kepler's Second Law:

The length of an arc is given by: $A_{sector} = \frac{1}{2}rl$

Hence: $A_{sector} = \frac{1}{2}rl = \frac{1}{2}r^2\Delta\theta$

Can be expressed as: $A = \frac{1}{2}r^2\frac{\Delta\theta}{\Delta t}\Delta t = \frac{1}{2}r^2\omega\Delta t$

Angular momentum given by: $L = mr^2\omega = \text{constant}$, hence $r^2\omega = \frac{L}{m}$.

Thus: $A_{sector} = \frac{1}{2}r^2\omega\Delta t = \frac{L}{2m}\Delta t = \text{constant} \times \Delta t$

KEPLER'S THIRD LAW: LAW OF PERIODS

The ratio of the cube of a planet's average distance from the Sun to the square of the planet's period of orbit has the same value for all the planets.

$$\frac{r^3}{T^2} = K$$

Kepler concluded that the radius and period of a planet's orbit were related.

- Where $K = 1$ if the units are kept as AU and years.
- r is the average distance of the planet from the Sun (or satellite from a planet).
- T is the period of the orbit.

KEPLER'S THIRD LAW & NEWTON'S LAW OF UNIVERSAL GRAVITATION

Using Newton's Law of Universal Gravitation, Kepler's Third Law, K can be predicted in any situation.

Since gravitational force on the planet is equal to the centripetal force needed to keep it moving in a circle:

$$F_{\text{GRAVITATIONAL}} = F_{\text{CENTRIPETAL}}$$

$$\frac{GmM}{r^2} = \frac{mv^2}{r}$$

Substitute $v = \frac{2\pi r}{T}$ and rearrange to get $\frac{r^3}{T^2} = \frac{GM}{4\pi^2}$

Therefore Kepler's Third Law is: $\frac{r^3}{T^2} = \frac{GM}{4\pi^2} = K$ where G is the universal gravitational constant and M is the mass of the central body around which the objects are orbiting.

- $K = \frac{GM}{4\pi^2}$ is **independent** of the mass of the planets. Therefore, it is a **constant** for all objects orbiting a certain central mass M.
- The equation $\frac{r^3}{T^2} = \frac{GM}{4\pi^2}$ tells us that if comparing different satellites of the **same central body**, then the ratio $\frac{r^3}{T^2}$ has a **fixed value**. This means that the ratio $\frac{r^3}{T^2}$ must be the **same** for each satellite

NOTES

- An object with a larger mass provides the force that keeps the object with the smaller mass moving in orbit. In the case of our solar system, the Sun provides the force that keeps the planets moving in orbit.
- Once the distance of a planet is known, then its orbital period and velocity can be determined.

ENERGY OF ORBITS

OUTCOMES

- derive quantitatively and apply the concepts of gravitational force and gravitational potential energy in radial gravitational fields to a variety of situations, including but not limited to:  
- the concept of escape velocity $v_{esc} = \sqrt{\frac{2GM}{r}}$
- total potential energy of a planet or satellite in its orbit $U = -\frac{GMm}{r}$
- total energy of a planet or satellite in its orbit $U + K = -\frac{GMm}{2r}$
- energy changes that occur when satellites move between orbits (ACSPH096)
- Kepler's Laws of Planetary Motion (ACSPH101)

TOTAL ENERGY OF AN ORBIT

A satellite in orbit will both have **kinetic energy** and **potential energy**.

- Potential energy is: $U = -\frac{GMm}{r}$
- Kinetic energy is: $KE = \frac{GMm}{2r}$ using $v_{orbitalspeed} = \sqrt{\frac{GM}{r}}$
- Hence total energy is: $E_{total} = U + KE = -\frac{GMm}{2r}$

UTILIZING THE EARTH'S ROTATIONAL MOTION

Relative to the centre of the Earth, the **ground is moving due to the Earth's rotation** about its own axis. Hence, a satellite on the ground will have **kinetic energy due to the Earth's rotation**.

Rockets can be sent into orbit more **efficiently** by taking **advantage** of **Earth's rotational motion**. To do this:

- It must be **launched near the equator** (to maximise its initial kinetic energy)
- Launched **towards the east**, so that the work done by the rocket is in the same velocity as the velocity of the ground.

ENERGY REQUIRED TO CHANGE ORBITS

The work required to move between orbits is the difference in energy of the two orbits: $W = E_f - E_i$. Hence:

$$W = \frac{GMm}{2} \left(\frac{1}{r_i} - \frac{1}{r_f} \right) = -\frac{GMm}{2} \left(\frac{1}{r_f} - \frac{1}{r_i} \right)$$

DIFFERENCE BETWEEN CHANGING ORBITS AND CHANGING POSITION

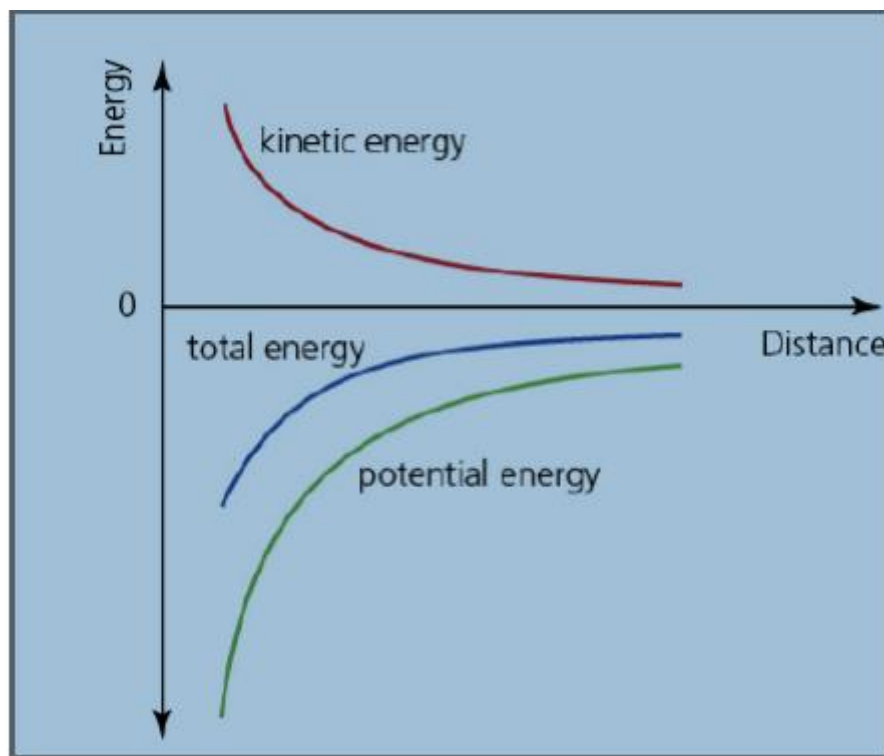
When moving from **one point in a gravitational field to another** it is assumed the object is stationary, so the **kinetic energy is zero**.

- Hence there will only be a change in potential energy for the object.
- The work required must provide the potential energy.

When moving **from one orbit to another** the right orbital velocity must be achieved for each orbit.

- Hence, both kinetic and potential energy will change.
- **Moving from an orbit close to Earth to an orbit further away from the Earth** will require an **increase in potential energy** but **decrease in kinetic energy**.
- Some of the **initial kinetic energy** can be **converted into potential energy**, so only half of the energy is required compared to simply moving from one point to another as above.

Change in orbit	Change in total energy	Energy transformations
Moving to a higher altitude orbit	Lower kinetic energy Higher potential energy Total energy increases	Some kinetic energy is converted to potential energy Additional positive work is required to increase total energy
Moving to a lower altitude orbit	Higher kinetic energy Lower potential energy Total energy decreases	Some potential energy is converted into kinetic energy Additionally negative work is required to decrease total energy



- Total energy is always negative
- With increasing distance, the KE decreases, and the GPE increases, as the total energy.

ESCAPE VELOCITY

NEWTON'S CONCEPT OF ESCAPE VELOCITY

Escape velocity is the **minimum velocity** needed at a given distance from the planet to **escape from the gravitational field** of that planet.

- Newton suggested that if a ball was projected horizontally it would travel a short distance before reaching the ground. If it was projected at a higher initial horizontal velocity, it would travel further before touching the ground.
- Newton then suggested if the ball was projected with a sufficiently high speed, it would fall towards the Earth at a slower rate than the Earth curves away from it. Therefore, it would never reach the Earth's surface and escape the Earth's pull. The velocity required to achieve this is called **escape velocity**.

ESCAPE VELOCITY FORMULA

The law of conservation of energy to find the minimum value of the initial speed such that an object will escape Earth's gravitational field.

- Total mechanical energy (KE and GPE in the system remains constant)

$$(E_K)_i + (E_P)_i = (E_K)_f + (E_P)_f$$

$$\frac{1}{2}mv_i^2 - \frac{GM_E m}{r_i} = \frac{1}{2}mv_f^2 - \frac{GM_E m}{r_f}$$

$$\frac{1}{2}mv_{esc}^2 = \frac{GM_E m}{r_E}$$

$$v_{esc} = \sqrt{\frac{2GM_E}{r_E}}$$

- The **starting conditions** are that the object is at the surface of the Earth, with initial velocity equal to escape velocity:
 - $r_i = r_e$ and $v_i = v_{esc}$
- The **final condition** is that the object has escaped out to infinitely far away from Earth:
 - $r_f = \infty$ and $v_f = 0$
- If the object is given an initial speed equal to v_{esc} , its total mechanical energy is equal to zero.

Escape velocity is:

- **Independent** of the **mass of the object** projected from the Earth. For example, a spacecraft has the **same** escape velocity as a molecule.
- **Independent** of the **direction of the velocity**, provided the trajectory does not intersect the Earth.

EQUATIONS

PROJECTILE MOTION

$$v_x = u_x$$

$$\Delta x = u_x t$$

$$v_y^2 = u_y^2 + 2a_y \Delta y$$

$$\Delta y = u_y t + \frac{1}{2} a_y t^2$$

$$v_y = u_y + a_y t$$

UNIFORM CIRCULAR MOTION

$$T = \frac{1}{f}$$

$$\omega = \frac{\Delta \theta}{\Delta t}$$

$$a_c = \frac{v^2}{r} = r\omega^2$$

$$v_{design} = \sqrt{gr \tan \theta}$$

$$\omega = 2\pi f = \frac{2\pi}{T}$$

$$v = 2\pi r f = \frac{2\pi r}{T}$$

$$F_c = \frac{mv^2}{r} = mr\omega^2$$

$$\tau = r_{\perp} F = rF \sin \theta$$

GRAVITATIONAL STUFF

$$F = \frac{GMm}{r^2}$$

$$U = -\frac{GMm}{r}$$

$$V_{orbital \ speed} = \frac{2\pi r}{T} = \sqrt{\frac{GM}{r}}$$

$$KE = \frac{1}{2}mv^2 = \frac{GMm}{2r}$$

$$W = \frac{GMm}{2} \left(\frac{1}{r_i} - \frac{1}{r_f} \right)$$

$$g = \frac{GM}{r^2}$$

$$\Delta U = -GMm \left(\frac{1}{r_f} - \frac{1}{r_i} \right)$$

$$\frac{r^3}{T^2} = \frac{GM}{4\pi^2} = K$$

$$E_{Total} = U + KE = -\frac{GMm}{2r}$$

$$v_{esc} = \sqrt{\frac{2GM}{r}}$$

Mass of Earth: $6.0 \times 10^{24} \text{ kg}$

Radius of Earth: $6.371 \times 10^6 \text{ m}$

Universal gravitational constant: $6.67 \times 10^{-11} \text{ Nm}^2\text{kg}^{-2}$

Position	Radial Force (Positive towards the centre)	Tangential Force (Positive in the same direction as velocity)
A: Bottom (Lowest Point)	$N - mg$	0
B: Right	N	$-mg$
C: Top (Highest Point)	$N + mg$	0
D: Left	N	$+mg$